TUSHAR

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GitHub ◇ LinkedIn

EDUCATION

Master's in Computer Science, New York University 2026 – 2028
Relevant Coursework: Design and Analysis of Algorithms II, Algorithmic Machine Learning and Data Science

Information Science Engineering, The National Institute of Engineering, Mysore 2020 – 2024
Relevant Coursework: Data Structures, Object-Oriented Programming in Java, Computer Networking, and Databases

EXPERIENCE

Independent Consultant, Abu Dhabi-based Client January 2026 – May 2026
Abu Dhabi, United Arab Emirates (Remote)

· Worked on full-stack development for a platform focused on the financialization of real-world assets.

Software Developer, Qdata, Sirius Holdings April 2025 – December 2025
Abu Dhabi, United Arab Emirates (On-site)

· Built a **CNF generator and Bitcoin mining implementation (SHA-256) from scratch**, optimizing for distributed execution.

· Created responsive trading dashboards on top of MT5 and its extensions.

· Architected a distributed CORBA/OmniORB manager–worker system for large-scale problem solving and Dockerized manager–worker services on AWS ECS, automating Terraform provisioning to scale workloads to **1,000+ concurrent workers**.

Backend and Data Engineer, Zator December 2024 – March 2025
New York, USA (Remote)

· Developed data streaming pipelines in Golang and Rust, fetching JSON/Protobuf data from APIs and Solana RPC.

· Engineered Kafka workflows (producers, consumers, topics) for efficient data filtering.

Young Research Fellow, IIIT Hyderabad IHub Foundation July 2024 – November 2024
Hyderabad, India (On-site)

· Engineered a real-time IoT system by collecting sensor data with MQTT, Fastify, and SQLite, and integrating it with machine learning insights for interactive visualizations.

· Researched and implemented on-device ML inference in React Native using ML Kit via a custom wrapper, TensorFlow Lite (react-native-fast-tflite), and ONNX (onnx-runtime-react-native).

PROJECTS

Monkey Compiler

· Developed in Go with no external dependencies; interprets integer and boolean types.

· Supports function calls, closures, and conditional expressions.

Market Maker Simulator

· Built a high-performance **Rust** market-maker simulator for ETH/USDC with live **Binance WebSocket**, **Jupiter** and **CoW API** price aggregation, custom execution logic, synthetic bid/ask modeling, and real-time PnL tracking.

· Designed and benchmarked a 10-minute automated quoting engine with a 2-second tick rate, a $100k notional target per session, dynamic fill-probability interpolation, a median-pricing model, and resilient async rate-limits and reconnection handling using the **Rust** async runtime and **cargo** suite.

ACHIEVEMENTS

· **First-batch solver** — Jane Street *Dropped a Neural Net*: 97-layer reconstruction (Hungarian pairing + hill climbing), exact match to historical predictions (MSE 0).